

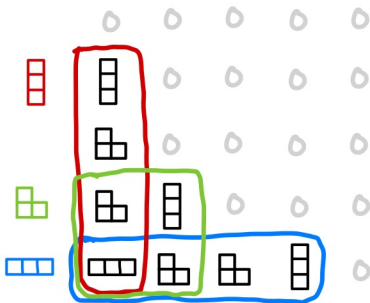
A Proof of Butler's Positivity Conjecture

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Section 1

Butler's Positivity Conjecture



Macdonald Polynomials

Schur [1901]

$$s_\lambda(X) \in \Lambda$$

Schur function



Littlewood [1961]

$$P_\lambda(X; t) \in \Lambda_t$$

Hall–Littlewood
symmetric function



Macdonald [1988]

$$P_\lambda(X; q, t) \in \Lambda_{q,t}$$

Macdonald
symmetric function

There are different forms of
Macdonald functions

$$P_\lambda = m_\lambda + (\text{lower term})$$

$$J_\lambda = P_\lambda \cdot \prod_{\square \in \lambda} (1 - q^{a(\square)} t^{\ell(\square)+1})$$

$$H_\lambda = J_\lambda \left[\frac{X}{1-t} \right] = J_\lambda \begin{pmatrix} x_1 & tx_1 & t^2x_1 & \dots \\ x_2 & tx_2 & t^2x_2 & \dots \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

$$\tilde{H}_\lambda = t^{n(\lambda)} H_\lambda|_{t \mapsto t^{-1}}$$

Theorem (Haiman, 2001)

$$\tilde{H}_\lambda \in \sum \mathbb{N}[q, t] \cdot s_\theta.$$

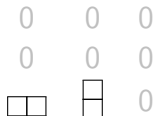
This was conjectured by Macdonald.

Example

$$\tilde{H}_{(1)} = s_{(1)}$$



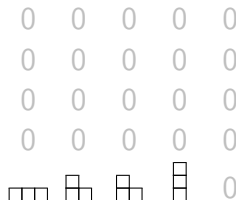
$$\tilde{H}_{(2)} = s_{(2)} + qs_{(1,1)}$$



$$\tilde{H}_{(2)} = s_{(2)} + ts_{(1,1)}$$



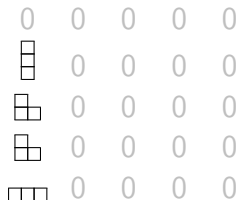
$$\tilde{H}_{(3)}$$



$$\tilde{H}_{(2,1)}$$



$$\tilde{H}_{(1,1,1)}$$



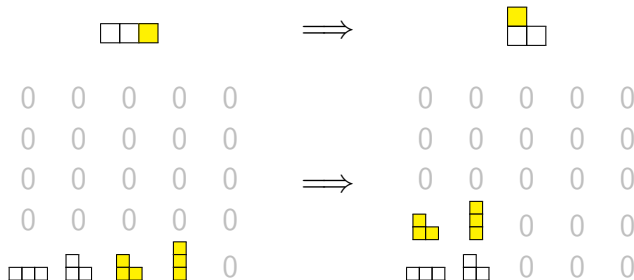
Butler's conjecture

Observing from the computation, Butler conjectured that

if μ is obtained by moving a box of λ , then

$\tilde{H}_\mu$ is obtained by moving a summand of $\tilde{H}_\lambda$.

For example,



Butler's conjecture

Note that if $\mu \vdash n$,

$$\tilde{H}_\mu = 1 \cdot h_n + \cdots + T_\mu \cdot e_n,$$

where $T_\mu = \prod_{(i,j) \in D(\mu)} q^i t^j = q^{n(\mu')} t^{n(\mu)}$. The precise conjecture can be stated as follows. we can write

$$\begin{aligned}\tilde{H}_\lambda(X; q, t) &= 1 \cdot I_{\lambda, \mu}(X; q, t) + T_\lambda \cdot J_{\lambda, \mu}(X; q, t) \\ \tilde{H}_\mu(X; q, t) &= 1 \cdot I_{\lambda, \mu}(X; q, t) + T_\mu \cdot J_{\lambda, \mu}(X; q, t),\end{aligned}$$

such that

$$I_{\lambda, \mu}(X; q, t), J_{\lambda, \mu}(X; q, t) \in \sum \mathbb{N}[q, t] \cdot s_\theta(X).$$

Butler's conjecture

Since $T_\lambda \neq T_\mu$, we have

$$l_{\lambda,\mu}(X; q, t) = \frac{T_\lambda \tilde{H}_\mu(X; q, t) - T_\mu \tilde{H}_\lambda(X; q, t)}{T_\lambda - T_\mu}. \quad (*)$$

Using the reciprocity identity:

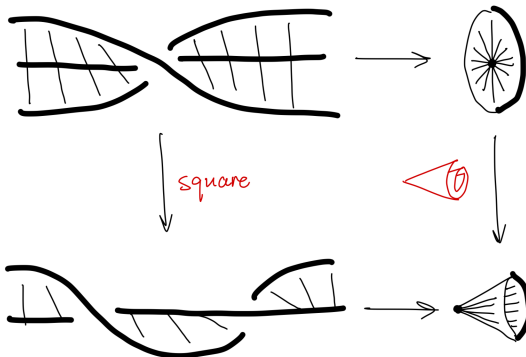
$$\tilde{H}_\mu(X; q, t) = t^{n(\mu)} q^{n(\mu')} \omega \tilde{H}_\mu(X; q^{-1}, t^{-1}),$$

it is easy to see $J_{\lambda,\mu}(X; q, t) = \omega l_{\lambda,\mu}(X; q^{-1}, t^{-1})$. So the conjecture is equivalent to

$$(*) \in \sum \mathbb{N}[q, t] \cdot s_\theta(X).$$

Section 2

Hilbert Schemes



Geometry of Hilbert schemes

Consider the **Hilbert scheme**

$$\begin{aligned} H_n &= \text{Hilbert scheme of } n\text{-points in } \mathbb{C}^2 \\ &= \{I \trianglelefteq \mathbb{C}[x, y] : \dim_{\mathbb{C}} \mathbb{C}[x, y]/I = n\}. \end{aligned}$$

By Fogarty, H_n is a nonsingular variety of dimension $2n$.

Also consider the **symmetric power**

$$\begin{aligned} S^n \mathbb{C}^2 &= (\mathbb{C}^2)^n / S_n = \operatorname{Spec} \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]^{S_n} \\ &= \{\text{multi-sets of } n\text{-points in } \mathbb{C}^2\}. \end{aligned}$$

We have the **Hilbert–Chow** map $\sigma : H_n \longrightarrow S^n \mathbb{C}^2$ by sending I to the support (with multiplicity) of $\mathbb{C}[x, y]/I$.

Geometry of Hilbert schemes

The **isospectral Hilbert scheme** X_n is defined to be the reduced fiber product in the following diagram

$$\begin{array}{ccc} X_n & \longrightarrow & \mathbb{C}^{2n} \\ \rho \downarrow & & \downarrow \\ H_n & \xrightarrow{\sigma} & S^n \mathbb{C}^{2n}. \end{array}$$

Theorem (Haiman)

The map ρ is finite and flat.

The **Procesi bundle** is the sheaf

$$\mathcal{P} = \rho_* \mathcal{O}_{X_n}.$$

This is a vector bundle of rank $n!$ by Haiman's Theorem!

Example ($n = 1$)

When $n = 1$, all spaces are $\mathbb{C}^2$, and all morphisms are identity.

$$\begin{array}{ccc} \mathbb{C}^2 = X_n & \longrightarrow & \mathbb{C}^{2n} = \mathbb{C}^2 \\ \rho \downarrow & & \downarrow \\ \mathbb{C}^2 = H_n & \xrightarrow{\sigma} & S^n \mathbb{C}^2 = \mathbb{C}^2 \end{array}$$

Any $I \in H_n$ is a maximal ideal, i.e. it takes the form $\langle x - x_0, y - y_0 \rangle$ for some $(x_0, y_0) \in \mathbb{C}^2$. So $H_n = \mathbb{C}^2$.

Since S_n is trivial, all others are also $\mathbb{C}^2$.

Example ($n = 2$)

When $n = 2$, we can identify H_n with the variety of

- ▶ two distinct points in $\mathbb{C}^2$ (**unordered**);
- ▶ a point with a tangent direction in $\mathbb{C}^2$.

We can identify $H_n = \mathbb{C}^2 \times \text{total space of } \mathcal{O}_{\mathbb{P}^1}(-2)$.

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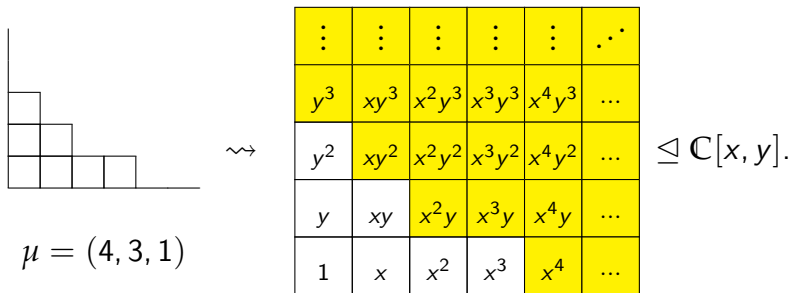
- ▶ two distinct points in $\mathbb{C}^2$ (**ordered**);
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We can identify $X_n = \mathbb{C}^2 \times \text{total space of } \mathcal{O}_{\mathbb{P}^1}(-1)$.

The map $\rho : X_n \rightarrow H_n$ is by taking square, which is flat.

Geometry of Hilbert schemes

For each partition $\mu \vdash n$, we can define a monomial ideal $I_\mu \in H_n$ as follows.



The fiber of Procesi bundle $\mathcal{P}$ at l_μ is a bigraded S_n -module. By Haiman, the Frobenius character is

$$\mathrm{Frob}_{q,t}(\mathcal{P}|_{I_\mu}) = \tilde{H}_\mu(X; q, t).$$

Section 3

Birkoff–Grothendieck Theorem

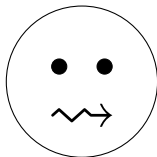
linear algebra
over field $\mathbb{C}$



linear algebra
over a curve

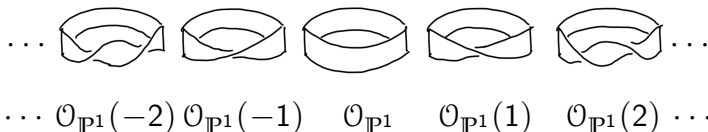


linear algebra
over a surface



Birkoff–Grothendieck Theorem

The line bundles (vector bundles of rank 1) over $\mathbb{P}^1$ are parametrized by $\mathbb{Z}$



For higher ranks, we have the following decomposition.

Theorem (Birkoff–Grothendieck)

Any vector bundle over $\mathbb{P}^1$ can be decomposed into a direct sum of line bundles.

Compare

over the field $\mathbb{C}$

A **vector space** can be decomposed into **lines** (**vector spaces** of **dimension** 1).

V.S.

over $\mathbb{P}^1$

A **vector bundle** can be decomposed into **line bundles** (**vector bundles** of **rank** 1).

More generally, let G be a reductive group. We have

over the field $\mathbb{C}$

A **G -representation** can be decomposed into **irreps**.

V.S.

over $\mathbb{P}^1$

A **G -equivariant vector bundle** can be decomposed into **???**.

Harder–Narasimhan Filtration

In the decomposition, the multiplicity of $\mathcal{O}_{\mathbb{P}^1}(i)$ is unique.
But the decomposition is **NOT canonical**.

What is canonical is the subbundle

$$\sum_{j \geq i} \left(\begin{array}{c} \text{summands} \\ \cong \mathcal{O}_{\mathbb{P}^1}(j) \end{array} \right) \quad (\text{direct sum}).$$

This forms the **Harder–Narasimhan filtration** of the vector bundle. Actually, it is the subsheaf generated by the image

$$H^0(\mathbb{P}^1, \mathcal{E} \otimes \mathcal{O}_{\mathbb{P}^1}(-i)) \otimes \mathcal{O}_{\mathbb{P}^1}(i) \longrightarrow \mathcal{E}.$$

Equivariant Birkoff–Grothendieck Theorem

Assume

$\mathcal{O}_{\mathbb{P}^1}(1)$ can be equipped with
a structure of G -equivariant line bundle.

(*)

What we have discussed shows the following

Theorem

Under (), any G -equivariant vector bundle over $\mathbb{P}^1$ can be decomposed into a direct sum of $\mathcal{O}_{\mathbb{P}^1}(i) \otimes V$ for $i \in \mathbb{Z}$ and V an irrep.*

- ▶ Here we can choose an equivariant splitting since $(-)^G$ preserves surjectivity.
- ▶ The condition (*) is true for any torus and GL_2 , but not for PGL_2 .

Return to Hilbert Schemes

Recall

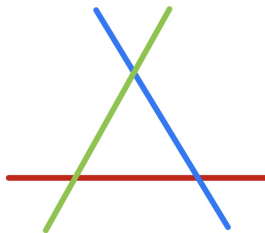
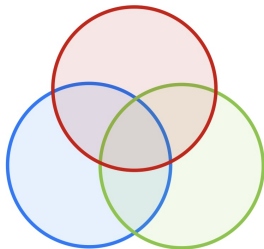
$$\begin{array}{ccc}
 X_n = \mathbb{C}^2 \times \left(\begin{array}{c} \text{total space} \\ \text{of } \mathcal{O}_{\mathbb{P}^1}(-1) \end{array} \right) & & \\
 \downarrow & \text{square} \downarrow & \\
 H_n = \mathbb{C}^2 \times \left(\begin{array}{c} \text{total space} \\ \text{of } \mathcal{O}_{\mathbb{P}^1}(-2) \end{array} \right) & \xleftarrow{\supset} & \mathbb{P}^1
 \end{array}$$

When restricting to $\mathbb{P}^1$, the Procesi bundle decomposed into

$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes \text{tri} \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes \text{sgn}.$$

Section 4

The Proof and Further



More Properties

To proceed the proof, we need two properties, more or less following from Haiman's proof.

The Procesi bundle $\mathcal{P}$ is generated by global sections.

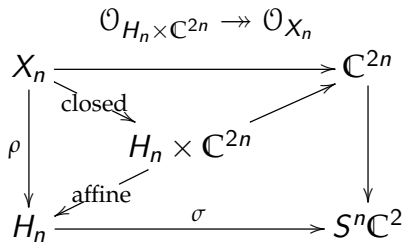
We have

$$\mathcal{P} \xrightarrow{\sim} \mathcal{P}^\vee \otimes \mathcal{O}_{H_n}(1) \otimes \mathbf{sgn}.$$

This is the geometric version of the reciprocity identity

$$\tilde{H}_\mu = T_\mu \cdot \omega \tilde{H}_\mu \Big|_{\substack{q \mapsto q^{-1} \\ t \mapsto t^{-1}}}.$$

Recall $T_\mu = t^{n(\mu)} q^{n(\mu')}$.



The Proof

Assume that μ is obtained by moving a box of λ .

Geometrically, we can connect the two points $l_\lambda, l_\mu \in H_n$ by a T -equivariant curve $\mathbb{P}^1$. Restricting to this $\mathbb{P}^1$, one can show the Procesi bundle splits into

$$\mathcal{P}|_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1} \otimes M \oplus \mathcal{O}_{\mathbb{P}^1}(1) \otimes N$$

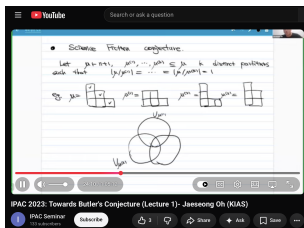
for bigraded S_n -representations M, N . Localizing at l_λ and l_μ , we get

$$\tilde{H}_\lambda = 1 \cdot \mathrm{Frob}_{q,t}(M) + T_\lambda \cdot \mathrm{Frob}_{q,t}(N),$$

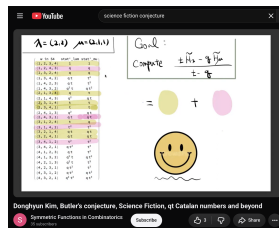
$$\tilde{H}_\mu = 1 \cdot \mathrm{Frob}_{q,t}(M) + T_\mu \cdot \mathrm{Frob}_{q,t}(N).$$

We are done.

Science Fiction Conjecture



talk by Jaeseong Oh



talk by Donghyun Kim

Conjecture (Part of Science Fiction Conjecture)

Let $\lambda \vdash n+1$ and $\mu(1), \dots, \mu(k)$ with $|\lambda/\mu(i)| = 1$. Then

$$\sum_{i=1}^k \prod_{j \neq i} \frac{T_{\mu(i)}}{T_{\mu(i)} - T_{\mu(j)}} \tilde{H}_{\mu(i)}(X; q; t) \in \sum \mathbb{N}[q, t] s_{\theta}(X).$$

Cohomological Vanishing Conjecture

Similarly, there is an embedding

$$f : \mathbb{P}^{k-1} \longrightarrow H_n$$

such that the T -fixed points are sent to $l_{\mu(1)}, \dots, l_{\mu(k)}$. By Atiyah–Bott localization theorem, we can show

Proposition

$$\left(\begin{array}{l} \text{the expression} \\ \text{in the conjecture} \end{array} \right) = \sum_{i \geq 0} (-1)^i \text{Frob}_{q,t} (H^i(\mathbb{P}^{k-1}, f^* \mathcal{P}^\vee)^*).$$

As a result, the conjecture can be implied by the following vanishing conjecture

$$\text{Conjecture} \quad H^i(\mathbb{P}^{k-1}, f^* \mathcal{P}^\vee) = 0, \quad \text{for any } i > 0.$$

Thanks